

Integrating Worldwide University Data from Wikidata, DBpedia, ROR and THE

Project Abstract – Web Data Integration, HWS 2026

Laith Sandouk, Mohammad Hadad, Sanjeevani Rajpurohit, Chellapuram
Keerthan Reddy, and Lakpa Dolma Sherpa

University of Mannheim, Germany

1 Use Case

Students, researchers and other stakeholders who need information about universities worldwide face a fragmented landscape: registries, knowledge bases and rankings differ in format, definitions and coverage, which makes institutions hard to compare. The Research Organization Registry (ROR) has reliable names and locations, but few descriptive facts. Wikidata and DBpedia add founding years, student numbers and institution types, but they are incomplete and often contradict each other. The Times Higher Education (THE) World University Rankings add scores, student statistics and subjects, but for only 3,100 institutions.

We integrate these four sources into one coherent dataset that describes every institution exactly once. Students can compare universities by location, ranking and subjects; analysts, policy makers and university administrators can benchmark institutions without consolidating sources by hand.

2 Datasets: Schema and Basic Profile

Table 1 profiles the four datasets, one more than required. Each contains a single class, *University*, with one record per entity.

Data sources and preprocessing. Our Python scripts retrieved all data on 2 October 2026 and store each source as one CSV file with one row per university. Attributes with several values are kept as lists, and website links that differ only in `http/https` or `www` are merged.

- *Wikidata*, <https://query.wikidata.org/sparql>: all items that are an instance of *university* (Q3918) or of one of its subclasses. Identifiers of countries, places and types were replaced by English labels; per university we kept the earliest founding year, the preferred or most recent student number and the city of its headquarters or location.
- *DBpedia*, <https://dbpedia.org/sparql>: all resources of class `dbo:University`, fetched with one paged query per attribute, because the endpoint returns at most 10,000 rows. We kept the earliest founding year and the largest student number, cut leftover wiki markup off the website links and unwrapped archived links.

- *ROR*, data dump v2.13 of 22 September 2026, <https://doi.org/10.5281/zenodo.22902037>: of its 141,528 records we kept the 30,131 of type *education* that are not withdrawn (withdrawn records are merged duplicates).
- *THE*, <https://www.timeshighereducation.com/world-university-rankings/2026/world-ranking>: the table that the web page loads, with 2,191 ranked institutions and 927 “reporters” that supplied data but got no rank. We added the ranking status, turned text such as “22,005” and “43%” into numbers and split the subjects into a list.

Table 1. Datasets

Dataset	Source (*)	Format	# of entities	# of attr.	List of attributes (**)
Wikidata	SPARQL endpoint	CSV	22,697	16	Wikidata id, name, alt. names (MV), acronyms (MV), country, country code, city, website, founding year, latitude (MV), longitude (MV), students (MV), types, member of (MV), ROR id (MV), Wikipedia title (MV)
DBpedia	SPARQL endpoint	CSV	27,728	14	DBpedia id, name, alt. names (MV), country, state (MV), city, website, founding year (MV), latitude (MV), longitude (MV), students (MV), types (MV), affiliations (MV), Wikidata id
ROR	Data dump (Zenodo)	CSV	30,131	18	ROR id, name, alt. names, acronyms (MV), country, country code, region, city, website, domains (MV), founding year, latitude, longitude, types, status, parent organisation (MV), Wikidata id, Wikipedia title (MV)
THE	Ranking web page	CSV	3,118	22	THE id, name, country, ranking status, rank, overall score, 5 pillar scores, 5 pillar ranks, students, students per staff, international students %, female students %, subjects, profile URL

(*) See “Data sources and preprocessing”. (**) (MV) = more than 30% missing values. The identifiers and the links to other sources serve only as gold standard for the matching.

Missing values. Over all attributes, 32% of the values are missing in Wikidata, 33% in DBpedia, 17% in ROR and 17% in THE. This does not hinder identity resolution, because the attributes we match on are well filled: the name is never missing and the country only in 3% of the Wikidata and 19% of the DBpedia

records. For the fusion the gaps are the reason for the project, because the sources fill each other’s gaps: student numbers are missing in 84% of the Wikidata and 67% of the DBpedia records but in only 1% of the THE records, and the coordinates missing in 31% of the Wikidata records are complete in ROR.

Ranking coverage. The largest gap is the ranking: THE lists only 3,118 institutions, 6% of the joint dataset (2,191 with a rank and 927 reporters without one). We see two options and would welcome feedback: (a) keep all institutions and label those that THE does not list as *unranked* in the ranking status; or (b) restrict the final dataset to the institutions that THE lists. Option (b) makes the dataset dense and, with 3,118 entities, still meets the required 2,500, but it stays below the range of more than 10,000 that the requirements call good.

3 Integrated Schema and Overlap with Input Schemata

The integrated schema has one class, *University*, with 34 attributes (Table 2). 14 attributes occur in at least two datasets and nine in at least three, eight of them besides the name, so fusion by voting is possible. Six attributes are lists. No single attribute serves as a key: the identifiers of the sources are not part of the schema, and the name alone is ambiguous (in ROR, “National University” exists in four countries). A university is identified by the combination of name and country, which is unique for 97.2% of the Wikidata, 99.8% of the DBpedia and ROR, and all THE records; the city separates most of the rest.

Table 2. Attribute Intersection with Integrated Schema

Attribute name	Attribute type	Datasets in which the attribute is found
name, country	string	Wikidata, DBpedia, ROR, THE
alternative names	list of strings	Wikidata, DBpedia, ROR
city	string	Wikidata, DBpedia, ROR
website	URL	Wikidata, DBpedia, ROR
founding year	date (year)	Wikidata, DBpedia, ROR
latitude, longitude	decimal	Wikidata, DBpedia, ROR
number of students	integer	Wikidata, DBpedia, THE
acronyms	list of strings	Wikidata, ROR
country code (ISO)	string	Wikidata, ROR
region (DBpedia: state)	string	DBpedia, ROR
institution types, memberships	list of strings	Wikidata, DBpedia
parent organisation, status	string	ROR
internet domains	list of strings	ROR
ranking status, rank, overall score	string	THE
5 pillar scores, 5 pillar ranks	decimal, integer	THE
students per staff, international	decimal	THE
students %, female students %		
subjects offered	list of strings	THE

The mapping is not one-to-one everywhere: the *affiliations* of DBpedia also contain parent universities, which we separate during translation, and the *types* of ROR only hold registry categories such as *education* or *funder*, so they are not mapped to the institution types.

4 Entities Contained in Multiple Datasets

Wikidata, DBpedia and ROR. These three sources reference each other through ROR identifiers, Wikidata identifiers and Wikipedia article titles (the DBpedia identifiers); 59% to 93% of the records of each source carry such a link. Following these links through the complete files, instead of a small sample, groups the records into 50,036 distinct institutions; 19,895 of them are contained in at least two sources and 9,829 in all three.

THE. THE shares no identifier with the other sources. Comparing normalised names with the names and alternative names of the other sources links 2,626 of the 3,118 THE universities (84%). This is a lower bound: of a random sample of 100 that we checked by hand, 98 are contained in another source, 16 of them under a different spelling. With THE, the joint dataset describes about 50,500 institutions: 19,993 in at least two sources, 10,125 in at least three and 2,225 in all four (Table 3).

Identity resolution remains a real task. The identifier columns are removed from the matching input and serve only as gold standard; for THE it is labelled by hand. Names alone do not suffice: 25% of the linked Wikidata–ROR and 18% of the linked Wikidata–DBpedia pairs differ in name even after normalisation.

Table 3. Project requirements and how the project meets them

Requirement	Our project
1. 3 different datasets	4: Wikidata, DBpedia, ROR, THE
2. $\geq 2,500$ entities in total, good: $> 10,000$	about 50,500 (from 83,674 records)
3. $\geq 1,000$ entities in at least two datasets	about 20,000
4. ≥ 8 attributes, identifiable by two of them	34; name + country, no identifier
5. ≥ 5 attributes in two datasets, some in three	14 in two, 9 in three or more
6. At least one list attribute	6, e.g. alternative names, subjects

Independence of the sources. ROR is curated by its own team, and the THE figures are submitted by the universities themselves. DBpedia is extracted from the English Wikipedia, on which Wikidata partly draws as well, so we count the two as one source for name and coordinates (63% of the linked pairs have identical coordinates). For the attributes fused by voting they behave independently: where exactly one of the three linked sources deviates in the founding year, Wikidata and DBpedia agree against ROR least often (5.6%, against 8.5% and 6.6% for the other pairings). Without DBpedia, only the country would remain in three datasets.